

RNA methods

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Notes

- Update something about document. Has to be removed before finished!

Abstract

This document outlines detailed procedures on RNA. It goes into methods on extraction and storage of cellular RNA; conversion to cDNA; running a PCR reaction; analyzing results on a Southern Blot gel.

Notice

The procedures here can be modified based on empirical and practical results. Of note, some of these steps can be omitted and skipped (*e.g.* no need to thaw if sample is not frozen). When moving liquids from one container to another, do so under the fumehood to avoid possible contamination from air. Please document each completed step (and skipped step) with appropriate data written down synchronously so as to collect information on procedures. Please take the time to do these things while performing the procedures, and do not eyeball numbers/values long after they are done. This is the most scientific and robust practice in a laboratory.

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Procedure

1. RNA extraction and storage

Critical: RNA is very unstable when isolated. As soon as it is pure in suspension, it is to be put on ice. RNA is stored in -80°C . When working around RNA (on the RNA bench), filtered tips for pipettes are used, and everything should be nuclease-free (e.g. DEPC-treated water, nuclease-free eppendorf tubes, etc.).

2. cDNA synthesis and storage

By using a reverse transcriptase, RNA from any extraction is converted to cDNA and semi-amplified using a number of PCR cycles.

The protocol is from [qScript cDNA Synthesis Kit \(Quantabio\)](#).

2.0 Required Materials

Equipment	Reagents	Bio. Ent.	Containers
Pipettor	qScript Reaction Mix (5X)	RNA extract, 500 ng x tube	Eppendorf Tube
Filtered Tips	Nuclease-free water	qScript Reverse Transcriptase (20X), 1 pack	PCR array tubes (0.2 mL each)
PCR machine (thermal cycler)			

2.1. cDNA synthesis

1. Thaw all frozen components, place all of them on ice.
2. Add the following to a 0.2mL PCR array tube sitting on ice for a final volume of 20 μL :
 - RNA (500ng), variable volume – depends on RNA solution concentration
 - Nuclease-free water, variable volume
 - qScript Reaction Mix (5X), 4 μL
 - qScript RT, 1 μL
3. Vortex gently, then centrifuge for 10 seconds (PCR-tube centrifuge).
4. Place PCR array tubes in a thermal cycler with the following program:
 - 1 cycle: 22 $^{\circ}\text{C}$, 5 min
 - 1 cycle: 42 $^{\circ}\text{C}$, 30 min
 - 1 cycle: 85 $^{\circ}\text{C}$, 5 min
 - 4 $^{\circ}\text{C}$ hold
5. (Optional) After completion of cDNA synthesis, use 1/5th to 1/10th of the first-strand reaction (2–4 μL) for PCR amplification. If desired, cDNA product can be diluted with 10 mM Tris-HCl (pH 8.0), 0.1 mM EDTA and stored at -20°C .

3. Primers

3.1 Primer design

Primer design is computationally done through the online web-tool [NCBI Primer-BLAST](#). The designed primers will have a desired T_m , GC content, and specificity for the gene of interest. Because of isoforms and SNPs and genetic sequence variability in databases, [NCBI Gene](#) is generally used as the reference gene database. From here, the desired gene is selected, and the transcript page is sought. On the transcript page, under “Analysis”, “Design Primers” is selected and primers can be designed with desired parameters (*e.g.* on exon-exon junctions).

3.2 Ordering primers

From Sigma-Aldrich, primer sequences are ordered, designed from NCBI Primer Design. The primers are named, and the names should include the targeted gene, organism if relevant, and other information to distinguish gene, purpose or characteristics (*e.g.* CCR2ab, used to detect isoforms A and B of CCR2; while CCR5ab cannot detect the isoform C of CCR5). The primer sequences are put in from 5′ – 3′ direction (*i.e.* from last nucleotide to first in the antisense primer). Other options are:

- 0.025 mole
- Desalt
- Dry

3.3 Primer stock preparation

Primers arrive with a datasheet specifying their properties (*e.g.* T_m , 2ndry tendency, GC%). The datasheet also contains required volume of DEPC-treated water to reach 100 μ M concentration. To prepare primer stocks, for each tube add double the respective specified volume to achieve a final concentration of 50 μ M. Store the primer in $-20C^\circ$ with other DNA. Primer stocks are used 1 μ L for each PCR reaction.

Important: Keep the datasheet. It contains important information for later use.

4. PCR

4.0 Required Materials

Equipment	Reagents	Bio. Ent.	Containers
Pipettor	qScript Reaction Mix (5X)	RNA extract, 500 ng x tube	Eppendorf Tube
Filtered Tips	Nuclease-free water	qScript Reverse Transcriptase (20X), 1 pack	PCR array tubes (0.2 mL each)
PCR machine (thermal cycler)			

4.1 Preparing PCR tubes

Add the following to each 0.2mL PCR tubes for a final volume of 20 μ L:

- PCR mix: 10 μ L

- cDNA: 2 μ L

This can be increased if the detection is poor (or the targeted RNA is sparsely transcribed)

- Forward primer: 1 μ L

Leave out if negative control.

- Reverse primer: 1 μ L

Leave out if negative control.

- DEPC-treated water: 6 – 8 μ L

Fill until total volume 20 μ L

4.2 Running PCR in a thermal cycler

For semi-quantitative PCR, the desired program is designed, and the tubes are placed in the thermal cycler.

Critical: PCR tubes, when placed in the thermal cycler, should be closed air-tight. Otherwise, during the cycles, the PCR solution will evaporate and the experiment will fail (at least in the open tube).

The thermal cycler has 4 total steps, for a total duration of [01:16:00-01:36:00]:

1. Initialization (95°C; 00:01:00)
2. PCR reaction; 35-45 cycles (01:10:00 - 01:30:00)

2a. Denaturation (95°C; 00:00:30)

2b. Priming (60 – 70°C; 00:01:00)

This temperature is variable, it should be lower than the minimum T_m of all of the primers.

2c. Elongation (72°C; 00:00:30 - 00:02:00)

Optimal temperature for the polymerase enzyme to use the annealed primers to elongate the chain. Elongation step can be longer (e.g. 2 minutes) if the expected PCR product is longer (e.g. 2000 bp). 30 seconds is good for products less than 1000 bp.

3. Wrap-up polymerase reaction (72°C - 00:05:00)
4. Hold (4°C)

5. Southern Blot

5.0 Required Materials

Equipment	Reagents	Bio. Ent.	Containers
Microwave	1.5g Agar Powder TAE x1	RNA extract, 500 ng x tube qScript Reverse Transcriptase (20X), 1 pack	50mL beaker PCR array tubes (0.2 mL each)
Electrophoresis Apparatus			

5.1 Preparing Agar Gel

Generally, 1.5% Agar Gel is prepared ($g/100mL$), but the ration can be varied based on the size of the DNA being run.

To prepare a solution of 1.5% Agar Gel solution, mix:

- 1.5g agar powder
- 100 mL TAE x1

Swirl to mix, and microwave

5.2 Setting up Southern blot

5.3 Running DNA on the gel

Remove later - formatting templates

Note: some note that will be inside a box.

Critical: Some important stuff.

color Ref protocols, section N. Dancing highlighted text

Negative Control and Positive Control: text

list of colors available: black, blue, brown, cyan, darkgray, gray, green, lightgray, lime, magenta, olive, orange, pink, purple, red, teal, violet, white, yellow.

For mathematical formulas, use this template:

$$F(x) = (OD_{260} - OD_{320}) \div (OD_x - OD_{320}), x \in [230, 320]$$